Jeffrey Wang

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Research Interests

My research centers on rethinking the primitive building blocks of deep architectures and how new designs change what a model can learn and at what cost. My first-author work, PolyNeXt (ICML 2026), replaces standard nonlinearities with activation-free polynomial (Hadamard-product) modules that match or exceed activation-based vision backbones at lower compute. I am extending this architecture-first program toward new attention and layer primitives and toward recurrence that lets models reason in a learned latent space rather than by stacking depth.

Education

University of Wisconsin–Madison — Madison, WI
Ph.D., Electrical & Computer Engineering (Focus: Machine Learning) — *Aug. 2026 – May 2031*

- Advised by Grigorios Chrysos; recipient of the **Distinguished Graduate Fellowship**.

University of Wisconsin–Madison — Madison, WI
B.S. Computer Science (Honors), B.S. Data Science — *Aug. 2022 – May 2026*

- **GPA: 4.0/4.0**; Dean's List (Fall 2022 – Spring 2025).
- Senior Honors Thesis in Computer Science: *Polynomial Vision Networks: Architecture and the Limits of Block-Recurrent Compression* (advisor: Grigorios Chrysos). [thesis]

Publications

- **Jeffrey Wang**, Jonathan Gregory, Grigorios G. Chrysos. *Activation-Free Backbones for Image Recognition: Polynomial Alternatives within MetaFormer-Style Vision Models*. Proceedings of the 43rd International Conference on Machine Learning (**ICML 2026**). [code & models]
- Min-Hsuan Yeh, **Jeffrey Wang**, Xuefeng Du, Seongheon Park, Leitian Tao, Shawn Im, Yixuan Li. *Challenges and Future Directions of Data-Centric AI Alignment*. Proceedings of the 42nd International Conference on Machine Learning (**ICML 2025**), Position Paper Track.

Research Experience

University of Wisconsin–Madison — Oct. 2024 – Present
Researcher, PI: Grigorios Chrysos

- First author on **PolyNeXt** (**ICML 2026**): a family of activation-free vision backbones replacing ReLU/GELU/softmax with Hadamard-product polynomial modules (PolyMLP, PolyConv, PolyAttn).
- Matched activation-based state-of-the-art (CAFormer) at equal size while staying fully activation-free: **84.3%** top-1 on ImageNet-1K at **26M** params, scaling to **85.2%**.
- Outperformed prior polynomial networks by **2–3%** top-1 at **<half the FLOPs**, with strong out-of-distribution robustness (ImageNet-C/A/R/Sketch) and ADE20K segmentation.
- Ran ImageNet-scale training/eval in PyTorch (DDP, bf16): ablations, robustness, segmentation transfer.
- Built fully-polynomial (BatchNorm) variants using only adds and multiplies, a step toward FHE inference.

University of Wisconsin–Madison — Jan. 2024 – Sept. 2024
Researcher, PI: Yixuan (Sharon) Li

- Second author on an **ICML 2025** position paper on data-centric AI alignment; designed and ran experiments quantifying how noise in RLHF preference data degrades reward models.
- Designed a noise-robust preference-learning objective (Generalized Binary Cross-Entropy) that improved training stability and accuracy over label-smoothing baselines.

Industry Experience

Amazon Web Services — June 2026 – Aug. 2026
Software Development Engineer Intern (Return) — *Seattle, WA*

- Building a CloudTrail event de-duplication recommendation system that selects optimal inclusion-key sets to maximize data reduction for IAM Access Analyzer.
- Designed to compress ∼**100 TB/hr** of event data by **98%+**, cutting downstream storage and analysis cost.
- Engineering the distributed pipeline with Step Functions, Lambda, and a PySpark Glue ETL over Parquet.

Amazon May 2025 – Aug. 2025
Software Development Engineer Intern *San Diego, CA*

- Built a high-throughput DynamoDB CDC to Apache Iceberg/S3 ingestion system (∼**100K events/s**) with PySpark/EMR and Kinesis for ML-ready feature stores.
- Delivered a reusable AWS CDK construct to onboard any DynamoDB table with standardized schemas and reconciliation checks, cutting integration time from **weeks to minutes**.
- Hardened reliability and security with CloudFormation/IAM; shipped in TypeScript, Java, and PySpark.

Correkt AI Aug. 2023 – Aug. 2024
Co-founder & Machine Learning Engineer *Santa Barbara, CA*

- Designed and launched a multimodal AI search engine with cited sources and images, serving **40K+ users**; helped secure funding from CNSI, Crescent Fund, and SBAA.
- Fine-tuned LLaMA-3-70B with RLHF; built a synthetic-data pipeline ranking millions of preference pairs at **<1%** the cost of human annotation, yielding **+10%** preference rate vs. competitors.
- Built web-scale retrieval with Elasticsearch and Groq: **800M+** indexed pages, ∼**300ms** avg query latency.

2Sigma School May 2023 – Aug. 2023
Software Engineer Intern *Remote*

- Fine-tuned LLM agents and NLP pipelines for personalized feedback and question generation (cut teacher prep **30%**, **+23%** satisfaction), with code-execution-based hallucination detection.

Projects

Evaluating Modality Bias in Multimodal LLMs Jan. 2026 – May 2026
Research Project [webpage] *Madison, WI*

- Built an ablation pipeline for modality bias in multimodal LLMs (Qwen2.5-Omni, Gemma-3n).
- Found models misreport their modality use and flip **69%** of correct answers under adversarial text injection.

Quantum-Inspired SQL Query Optimization Sept. 2025 – Dec. 2025
Google-Sponsored Software Engineering Capstone *Madison, WI*

- Built a quantum join-reordering pipeline (QAOA, VQE), modeling SQL optimization as a QUBO in Qiskit.
- Achieved **15% lower** average estimated cost vs. the PostgreSQL baseline plan on 5+ table TPC-H queries.
- Implemented SQL-to-QUBO compilation from relational query plans for quantum-annealing optimization.

Depth Efficiency of Recurrent Networks Jan. 2025 – May 2025
Research Project *Madison, WI*

- Extended depth-efficiency theory (why deeper nets are exponentially more expressive) to recurrent models.
- Proved a Tensor-Train network is RNN-equivalent with the same exponential rank separation, via measure theory.

Honors & Awards

- Distinguished Graduate Fellowship, University of Wisconsin–Madison (2026)
- Dean's List (Fall 2022 – Spring 2025), University of Wisconsin–Madison
- USACO Gold (2021)

Service

- Reviewer, ICLR 2026 Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy

Technical Skills

- **Machine Learning:** Deep Learning, Transformers, Computer Vision, NLP, RLHF, Fine-tuning, Neural Architecture Search, RAG, Distributed Training
- **Frameworks:** PyTorch, Hugging Face, DeepSpeed, NumPy, Pandas, Qiskit
- **Cloud / Data:** AWS (Lambda, Glue, EMR, Kinesis, DynamoDB, CDK), GCP, Spark, Apache Iceberg
- **Backend:** Django, Flask, Elasticsearch, MongoDB, REST APIs
- **Languages:** Python, C/C++, Java, TypeScript, JavaScript, SQL
- **Tools:** Docker, Git, Linux, Slurm, HTCondor, CI/CD, CloudFormation, Weights & Biases